

Air University, Islamabad
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EduMaestro

*A LangChain Multi-Agent Retrieval-Augmented Tutor
for Curriculum-Aligned Learning*

Course	Information Retrieval (IR)
Instructor	
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Domain	Method	Framework	Language
Educational AI	Multi-Agent RAG	LangChain	English

1. Introduction

EduMaestro is a multi-agent intelligent tutoring system built on LangChain that combines classical Information Retrieval (IR) techniques with modern large language model (LLM) synthesis to deliver curriculum-aligned, evidence-grounded answers for students. The core motivation is simple but critical: students using general-purpose LLMs for self-study receive fluent and confident responses that are not always grounded in authoritative curriculum documents, making hallucination a serious educational risk.

EduMaestro addresses this by treating educational question-answering as a multi-stage retrieval problem. Every generated answer is anchored in verifiable passages retrieved from a closed, curated corpus of curriculum-aligned documents. Students can trace every claim back to its source — making the system explainable and trustworthy by design.

1.1 Motivation

Two persistent problems define the current landscape of AI-assisted self-study:

- **Ungrounded LLM Outputs:** General-purpose models such as Gemini or GPT-4 generate explanations and worked solutions without citing sources. In mathematics education, a single hallucinated formula or an incorrect derivation step can plant a misconception that is difficult for a student to self-detect and correct.
- **Fragmented Resource Discovery:** High-quality, curriculum-aligned resources are spread across textbooks, board papers, exercise sets, and online platforms. Students lack tools that intelligently retrieve material matching their current Learning Objective (LO) rather than returning generic web results.

EduMaestro solves both problems simultaneously: retrieval grounds the answers, and the multi-agent pipeline ensures that what is retrieved is curriculum-relevant, properly ranked, and pedagogically structured.

1.2 Scope & Domain

To keep the project feasible within the six-week timeline, EduMaestro targets a focused domain:

Subject	Secondary Mathematics — Algebra and introductory Geometry
Grade Level	Grades 9 and 10
Curriculum	National-board-aligned Learning Objectives (Pakistan)
Language	English (monolingual corpus and interface)
Corpus Size	1,000 – 2,000 curated documents / passage chunks

This scope is narrow enough to enable quality annotation and meaningful evaluation within six weeks, yet rich enough to demonstrate every component of the full IR pipeline — retrieval, reranking, curriculum alignment, generation, and assessment.

1.3 Key Contributions

1. A curated, annotated English corpus of curriculum-aligned secondary mathematics resources with Learning Objective labels, evidence spans, and prerequisite graph edges.
2. A seven-agent LangChain orchestration pipeline combining BM25, dense retrieval, cross-encoder reranking, and controlled LLM synthesis — purpose-built for education.
3. A faithfulness-first generation protocol: strict evidence-only prompting and post-generation token-overlap verification to minimize hallucination.
4. An integrated pedagogical layer: LO-mapped retrieval, concept-graph-driven learning paths, and adaptive formative assessment in a single pipeline.
5. A human user study measuring pre/post learning gains against a standard LLM baseline.

2. Dataset

The EduMaestro dataset is the foundation of the entire IR pipeline. It is purpose-built for curriculum-aligned secondary mathematics and comprises five distinct annotation layers that collectively support retrieval evaluation, evidence selection evaluation, generation faithfulness checking, and pedagogical path planning.

2.1 Source Categories

Raw documents are collected from five source types to ensure coverage across explanation, practice, and assessment material:

Source Type	Specific Examples	Target Volume
Official Curriculum Docs	National board syllabi, LO frameworks, grade descriptors	50 – 80 documents
Open Textbooks	OpenStax Algebra & Geometry chapters, CK-12 textbook pages	200 – 300 chapters
Online Educational Content	Khan Academy lesson transcripts, Wikipedia math articles	400 – 600 articles
Past Exam Papers & Solutions	Board exam question papers with official solutions	100 – 150 papers
Teacher-Contributed Notes	Solved worked examples, rubric sheets, explanation snippets	80 – 120 documents

2.2 Annotation Layers

Five annotation layers are constructed on top of the raw document collection. Together they provide the gold-standard labels required to train and evaluate each stage of the pipeline.

Layer 1 — Learning Objective (LO) Inventory

150 to 300 explicit Learning Objectives are parsed from official curriculum documents and organized by topic cluster and Bloom's Taxonomy cognitive level (Recall, Apply, Analyze). Each LO is given a canonical ID used throughout all other layers.

Layer 2 — LO-to-Resource Relevance Labels

400 to 600 (LO, document chunk) pairs are labeled with graded relevance: 0 = irrelevant, 1 = partially relevant, 2 = highly relevant. These labels are used to train the cross-encoder reranker and to compute offline retrieval metrics (nDCG, MRR, Recall@K).

Layer 3 — QA Pairs with Evidence Spans

300 question-answer pairs are created across all topics. For approximately 200 of these, supporting evidence spans are manually annotated at the sentence level within retrieved documents. These span annotations are the gold standard for evidence selection evaluation (span F1).

Layer 4 — Concept Prerequisite Graph

150 directed prerequisite edges are validated from curriculum sequencing guidelines. Example edges: "Factoring Polynomials" must precede "Solving Quadratic Equations"; "Properties of Angles" must precede "Triangle Congruence". This graph powers the learning path recommendations in the Planner Agent.

Layer 5 — Document Metadata

Each document chunk is tagged with four metadata fields used for filtering and LTR features: document type (definition, worked example, exercise, solution), grade level (9 or 10), topic cluster, and estimated reading difficulty (Flesch-Kincaid score).

2.3 Processing Pipeline

Raw documents are transformed into indexed, searchable passage chunks through the following pipeline:

6. Extraction: pdfminer and PyMuPDF extract text from PDF textbooks; direct HTML parsing handles web resources.
7. Cleaning: Unicode normalization, removal of headers/footers/page numbers, preservation of LaTeX-formatted mathematical expressions.
8. Chunking: Sentence-aware chunking at 200–350 tokens per chunk with 50-token overlap between adjacent chunks to prevent context loss at boundaries.
9. Metadata tagging: doc_type, grade_level, topic_cluster, difficulty_score appended per chunk.
10. Dual indexing: Chunks are loaded into Elasticsearch for BM25 keyword search and into FAISS for dense vector search using sentence-transformer embeddings.

2.4 Dataset Summary

Artifact	Count
Raw documents / articles	1,000 – 2,000
Processed passage chunks	8,000 – 15,000 (est.)
Learning Objective entries	150 – 300
LO-resource relevance label pairs	400 – 600
QA pairs (total)	300
QA pairs with evidence spans	~200
Prerequisite graph edges	~150
Unique topic clusters	12 – 18

3. Problem Statement

EduMaestro formalizes educational question-answering as a retrieval-augmented generation problem with four simultaneous quality constraints. The formal definition is:

Formal Problem Definition

Given a student query Q (natural language, English) and an optional learner profile $P = \{ \text{grade level, set of mastered concept IDs} \}$, retrieve from a closed educational corpus C a ranked set of curriculum-aligned evidence passages R , and generate a grounded answer A such that: (i) every factual claim in A is supported by a passage in R with an inline citation, (ii) a step-by-step worked solution is provided where applicable, (iii) a 5–10 minute lesson outline is included, and (iv) 3 formative assessment questions with rubrics are appended. Additionally, recommend the next concept(s) along the prerequisite graph given P .

3.1 Quality Constraints

The pipeline must simultaneously satisfy the following four constraints. Each constraint maps directly to a measurable evaluation metric.

Constraint	Definition	Primary Metric
Faithfulness	Every factual claim in the generated answer must be traceable to a specific passage in the retrieved set R .	Faithfulness rate (%)

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Retrieval Relevance	Passages in R must be relevant to the student's current Learning Objective as determined by the gold-standard LO labels.	nDCG@5, nDCG@10, MRR
Pedagogical Utility	The output must be usable for independent self-study: correct step-by-step solutions, appropriate difficulty, and assessments.	Human expert grading, learning gain
Efficiency	End-to-end latency from query to complete response must be acceptable for a live interactive session.	Latency (seconds) < 8s target

3.2 Subtasks

The full problem decomposes into five subtasks, each addressed by a dedicated agent:

- Query Understanding: Map raw natural language input to a canonical query, intent type, and target Learning Objective ID(s).
- Curriculum-Aligned Retrieval: Retrieve the top-N passages from the corpus that are most relevant to the target LO using hybrid BM25 + dense retrieval.
- Evidence Reranking & Selection: Re-score and select the top-K evidence passages using a cross-encoder, ensuring both relevance and diversity.
- Evidence-Grounded Generation: Synthesize a structured, cited answer using only the selected evidence passages; refuse to generate beyond evidence.
- Pedagogical Scaffolding: Map the answer to the concept graph, identify prerequisite gaps, and generate formative assessment questions.

3.3 Out-of-Scope Constraints

The following are explicitly outside the scope of this project to maintain feasibility:

- Multilingual or cross-lingual retrieval — the system is English-only.
- Real-time learner modelling at scale — the learner model is a lightweight per-session Bayesian update, not a full persistent learning management system.
- Subjects beyond secondary mathematics — the corpus and LO inventory are scoped to Algebra and Geometry, Grades 9–10.

4. Literature Review

EduMaestro sits at the intersection of four active research areas: retrieval-augmented generation, dense retrieval and reranking, intelligent tutoring systems, and multi-agent LLM orchestration. This section reviews key works in each area and identifies the specific gaps that EduMaestro addresses.

4.1 Retrieval-Augmented Generation (RAG)

Lewis et al. (2020) introduced RAG as a framework that conditions language model generation on a set of retrieved documents, substantially improving factual accuracy over closed-book generation. RAG demonstrated that grounding answers in retrieved text is effective for open-domain question answering. However, the original formulation uses a single retrieval stage and a single generative model with no curriculum alignment or pedagogical structure.

Subsequent work such as REALM (Guu et al., 2020) and Atlas (Izacard et al., 2022) refined the retriever-generator interaction but remained focused on factoid QA benchmarks rather than educational settings. EduMaestro extends RAG with multi-stage retrieval, cross-encoder reranking, evidence selection, and a structured pedagogical output format — none of which appear in prior RAG pipelines.

4.2 Dense Retrieval & Hybrid Search

Dense Passage Retrieval (DPR, Karpukhin et al., 2020) established the bi-encoder paradigm for semantic document retrieval, encoding queries and passages independently and computing similarity via dot product. DPR outperforms BM25 on paraphrased and conceptual queries, which are common in student questions. However, DPR alone suffers on exact-match queries (theorem names, formula labels) where BM25 excels.

Hybrid retrieval with Reciprocal Rank Fusion (RRF, Cormack et al., 2009) combines the lexical precision of BM25 with the semantic coverage of dense retrieval without requiring score normalization. EduMaestro adopts this hybrid strategy in the Retrieval Agent to capture both query types effectively.

4.3 Cross-Encoder Reranking

Nogueira and Cho (2019) demonstrated that fine-tuned BERT cross-encoders significantly outperform bi-encoders on passage reranking tasks, at the cost of higher inference time. Because cross-encoders jointly encode the query and each candidate passage, they capture fine-grained query-passage interaction that bi-encoders cannot. EduMaestro uses a cross-encoder as the reranker (Agent 3), applied only to the top-50 BM25+dense candidates to control latency.

4.4 Hallucination & Faithfulness in LLM Generation

Maynez et al. (2020) and Shuster et al. (2021) documented that language models routinely generate plausible but unsupported factual claims (hallucinations), even when source documents are provided in context. FaithfulQA (Dziri et al., 2021) and FEVER-style evaluation frameworks provide benchmarks for measuring faithfulness. In educational settings the stakes are higher: a hallucinated derivation step in a worked solution can mislead a student in a way that is difficult to self-correct.

EduMaestro responds with two mechanisms: (1) an evidence-only prompt template that instructs Gemini to generate only from provided passages and to refuse generation if evidence is insufficient, and (2) a post-generation token-overlap faithfulness check that flags responses below a confidence threshold.

4.5 Intelligent Tutoring Systems (ITS)

Classical ITS research (VanLehn, 2011) identifies four core components: a domain model, a student model, a pedagogical model, and an interface. Systems such as ASSISTments, ALEKS, and Cognitive Tutor implemented these components with hand-crafted rule bases. More recent systems (Piech et al., 2015 — Deep Knowledge Tracing) replaced hand-crafted models with neural student models that predict concept mastery from response sequences.

EduMaestro adopts the ITS architectural philosophy but replaces rule-based domain and pedagogical models with retrieval-driven components: the corpus plus LO inventory form the domain model; the concept prerequisite graph is the pedagogical model; and a lightweight Bayesian Knowledge Tracing update serves as the student model. This makes EduMaestro domain-agnostic and scalable to any document corpus.

4.6 Multi-Agent LLM Orchestration

HuggingGPT (Shen et al., 2023) and ReAct (Yao et al., 2022) showed that decomposing complex tasks across specialized agents — rather than asking a single LLM to do everything in one pass — improves reliability and interpretability. LangChain (Chase, 2022) provides a framework for implementing such orchestration with standardized agent interfaces, tool definitions, and chain composition.

No prior work applies multi-agent LangChain orchestration to curriculum-aligned educational retrieval with specialized agents for query understanding, retrieval, reranking, alignment, generation, assessment, and coordination. EduMaestro fills this gap.

4.7 Gap Summary & Novelty

Prior Work	What It Covers	What EduMaestro Adds
Lewis et al. RAG (2020)	Single-stage retrieval + generation	Multi-stage pipeline, curriculum alignment, pedagogical output
DPR (Karpukhin 2020)	Dense retrieval for open-domain QA	Hybrid BM25+dense with RRF, education domain-specific
Nogueira & Cho (2019)	Cross-encoder reranking	Applied to LO-aligned evidence selection in education
DKT (Piech et al. 2015)	Neural knowledge tracing	Simplified BKT integrated with retrieval pipeline
HuggingGPT / ReAct	Multi-agent task decomposition	Applied to IR stages in an educational RAG setting
ASSISTments / Cognitive Tutor	Rule-based ITS with student modelling	Corpus-driven, scalable, LLM-synthesized explanations

5. Proposed Solution

EduMaestro is implemented as a LangChain multi-agent system. Seven specialized agents handle distinct stages of the IR pipeline, coordinated by a Controller Agent that sequences execution, manages caching, and handles fallback logic. The system exposes a REST API (FastAPI) consumed by a React/Next.js frontend.

5.1 High-Level Architecture

Agent	Role	Key Technology
1 — Query Understanding	Parse query intent, map to target LO(s)	Sentence Transformers + LO embedding index
2 — Retrieval	Hybrid BM25 + dense retrieval, top-50 candidates	Elasticsearch (BM25) + FAISS (dense)
3 — Reranker & Selector	Cross-encoder reranking, top-K evidence selection	Fine-tuned BERT CrossEncoder + MMR diversity
4 — Curriculum Planner	LO alignment check, prerequisite path generation	Concept prerequisite graph (NetworkX)
5 — Generator	Evidence-grounded structured answer synthesis	Gemini 1.5 Flash API (evidence-only prompt)
6 — Assessment	Formative question generation, learner model update	Gemini prompt + Bayesian Knowledge Tracing
7 — Controller	Pipeline orchestration, caching, fallback logic	LangChain AgentExecutor + Redis LRU cache

5.2 Agent Specifications

Agent 1 — Query Understanding Agent

Transforms raw student input into a structured representation consumed by all downstream agents. Detects query intent (factoid lookup, conceptual explanation, worked problem solving, or resource discovery) using a lightweight intent classifier. Maps the query to one or more target Learning Objective IDs by computing cosine similarity between the query embedding and all LO embeddings. Extracts mathematical entities such as variable names, operation types, and theorem labels. Output contract: { canonical_query, intent_type, target_LOs[], proficiency_level }.

Agent 2 — Retrieval Agent

Executes hybrid retrieval over both indices. BM25 search (Elasticsearch) handles exact-match queries — formula names, theorem labels, specific terminology. Dense search (FAISS + all-MiniLM-L6-v2 sentence embeddings) handles conceptual and paraphrased queries. Reciprocal Rank Fusion combines the two ranked lists into a single

unified ranking without requiring score normalization. Output: top-50 candidate passages with metadata, source document references, and per-passage BM25 / dense scores.

Agent 3 — Reranker & Evidence Selector Agent

Applies a fine-tuned BERT cross-encoder to score each (query, passage) pair from the top-50 candidates. The cross-encoder is fine-tuned on our LO-resource relevance labels and QA-evidence span labels, starting from an MS MARCO pre-trained checkpoint. Maximal Marginal Relevance (MMR) is applied after reranking to select a diverse top-K set (5–10 passages), avoiding redundant evidence. An evidence confidence score (normalized aggregate) triggers a fallback to the evidence list if it falls below threshold. Output: top-K evidence passages with citation IDs and confidence scores.

Agent 4 — Curriculum Alignment & Planner Agent

Verifies that each selected evidence passage maps to the student's target LO and filters off-topic material. Queries the prerequisite concept graph to check whether the student's mastered concept set covers the prerequisites for the target LO. If gaps are found, a prerequisite warning is included in the final response. Computes a recommended next-concept sequence (2–3 LOs) using shortest-path traversal on the graph. Output: filtered and LO-aligned evidence passages, prerequisite gap flags, recommended next LO sequence.

Agent 5 — Synthesis (Generator) Agent

Receives the filtered evidence passages and generates the final structured student-facing response using Gemini 1.5 Flash. The prompt template enforces three hard rules: (1) generate only from the supplied passages; (2) if passages are insufficient, respond with 'Insufficient evidence to answer from the curriculum corpus' rather than hallucinating; (3) cite every factual claim inline with a passage ID. The output is structured into four sections: Direct Answer with citations, Step-by-Step Worked Solution, 5-Minute Lesson Outline, and Three Formative Questions with Rubrics. A post-generation faithfulness check computes token overlap between the answer and evidence passages; responses below threshold are flagged.

Agent 6 — Assessment Agent

Generates three formative questions per query at varied Bloom's Taxonomy levels: recall (Level 1), application (Level 3), and analysis (Level 4). Each question includes a model reference answer and a simple scoring rubric. When the student submits responses, a Bayesian Knowledge Tracing update modifies the per-concept mastery probability stored in the session learner model. This mastery state is used by Agent 4 in subsequent queries to provide better-targeted prerequisite recommendations.

Agent 7 — Controller / Coordinator Agent

The LangChain AgentExecutor that sequences all agents. Execution order adapts to query intent: problem-solving queries route through all six agents in full; resource-discovery queries skip the Generator and Assessment agents, returning only the ranked evidence list. A Redis LRU cache stores (query hash → full response) for repeated queries. If the Gemini API is unavailable or evidence confidence is below threshold, the Controller falls back to returning the top-5 evidence passages with direct sentence excerpts as the answer.

5.3 Model & Training Summary

Component	Model / Approach	Training Signal
Dense Retriever	all-MiniLM-L6-v2 (Sentence Transformers)	Off-the-shelf; optional fine-tune on LO→resource pairs
Cross-Encoder Reranker	BERT-base CrossEncoder (MS MARCO checkpoint)	Fine-tune on LO-resource labels + QA evidence spans
LTR Feature Ranker	LightGBM on 6-feature vector	BM25 score, dense sim, cross-encoder score, doc type, difficulty, LO match
Generator	Gemini 1.5 Flash (API)	Prompt engineering only — no fine-tuning
Learner Model	Bayesian Knowledge Tracing (BKT)	Student formative response signals per session

5.4 Prompt Engineering Strategy

The Generator Agent (Gemini) is controlled through a rigid prompt template rather than fine-tuning, for cost and reproducibility. The template enforces the following structure in every call:

- System instruction: You are an educational tutor. Answer ONLY using the passages provided below. If the passages do not contain sufficient information to answer the question, respond exactly with: 'Insufficient evidence.'
- Evidence block: Numbered list of top-K passages with IDs, source document references, and metadata.
- Task specification: Answer the student query, cite passage IDs inline, then provide a step-by-step solution, a 5-minute lesson outline, and three formative questions with rubrics.
- Output format enforcement: JSON-structured response with keys: answer, citations, worked_solution, lesson_outline, formative_questions.

5.5 Evaluation Plan

Evaluation Axis	Metric	Target
LO-to-resource retrieval	nDCG@5, nDCG@10, MRR	> 0.70 / > 0.65
Retrieval coverage	Recall@10	> 0.80
Evidence selection	Span F1 vs. human annotations	> 0.60
Answer faithfulness	% claims supported by evidence	> 0.90
Generation quality	Human expert grading (accuracy, clarity)	Score $\geq$ 4 / 5
Learning gain (user study)	Pre/post test score delta	Statistically significant ($p < 0.05$)
System efficiency	End-to-end latency per query	< 8 seconds

An ablation study will compare four system configurations: (A) Gemini-only with no retrieval; (B) BM25-only RAG; (C) Hybrid RAG without cross-encoder reranking; and (D) EduMaestro full pipeline. This isolates the contribution of each component to the final performance.

5.6 Demo Architecture

The live demonstration runs a React/Next.js frontend connected to a FastAPI backend. The interface exposes five panels:

- Query Panel: Natural language input with detected intent badge.
- Evidence Panel: Top-5 retrieved passages with source references and highlighted supporting sentences.
- Answer Panel: Generated answer with inline citation superscripts and a per-claim faithfulness indicator.
- Lesson & Quiz Panel: 5-minute lesson outline and 3 formative questions with expandable rubrics, toggled via a button.
- Concept Graph Panel: D3.js visualization of the prerequisite graph with the current concept highlighted and recommended next steps indicated.

Three pre-baked demo scenarios are pre-computed and cached to ensure a smooth live presentation: (1) a worked problem query on quadratic equations, (2) a concept discovery query for prerequisite planning, and (3) an instructor resource-discovery query filtered by LO and grade level.

5.7 Project Timeline

Week	Focus	Deliverables
Week 1	Scope & Data Acquisition	Corpus collection scripts, text extraction pipeline, ~1,000 raw docs, canonical LO list
Week 2	Indexing & Baseline Retrieval	Elasticsearch + FAISS indices, BM25 baseline, 300 QA seed pairs, embedding generation
Week 3	Annotation & Reranker Training	400 LO-resource labels, 200 evidence spans, cross-encoder fine-tuning, LightGBM LTR
Week 4	Multi-Agent Orchestration	All 7 agents implemented in LangChain, Gemini prompt templates, caching, fallbacks
Week 5	Frontend & Integration	React/FastAPI integration, all UI panels, pre-baked demo scenarios, internal pilot
Week 6	Evaluation & Write-up	Human user study (20–30 participants), ablation results, final paper, demo video